Aidin Kazempour

Researcher

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About Me

◦ Graduate Student of Mechatronic Engineering who intensely passionate about Computer Science filed specially subject like Reinforcement Learning, Evolutionary Algorithms and AI with strong experience in implementing related Algorithm in each field.

◦ have spended considerable amount of time to learn computer science subject included datastructure and algorithms.

◦ familiar with methods of researching and dealing with papers. have being been learnt these skills through Master of Science Thesis.

◦ high skills in both doing projects alone or cooperating with other fellows as a team member and flexibility in adaptation to new environments

Education

M.Sc Mechatronic Engineering 01/08/2013 - 01/08/2017

University of Tabriz, Tabriz, IRAN

Thesis Topic: An Evolutionary Reinforcement Learning Algorithm

- Grades: AI : 19.5/20, DeepLearning: 19/20, Modern Control: 18/20, Advance Mathematic: 16.5/20
- GPA : 4.00/4.00

B.Sc Mechanical Engineering 01/08/2018 - 01/08/2021

University of Tabriz, Tabriz, IRAN

- Grades: Engineering Mathematic : 19.5/20, Differential Equation: 20/20, Programming: 18/20, Numerical Computation: 19/20, Statistics: 16/20, Linear Control: 15/20
- GPA : 3.00/4.00 (last 2 year : 3.4/4.00)

Projects

○ University of Tabriz

↪ AI Research Scientist

- An Evolutionary Reinforcement learning method in cooperative multi-agent-environments
  - implemented two state of art Reinforcement Learning method in multi-agent cooperative settings (QMIX, MADDPG) {preapring environments, creating buffers, building neural network models and etc.}
  - implemented couple of swarm intelligence methods like Ant Colony Optimization, Gray Wolf Optimization, Particle Swarm Optimization and etc.
  - combining Rl & Swarm Intelligence methods to deal with sparse rewards
  - ⋆ Under Supervision of dr. Behruz Kuhestani

○ Self Projects

↪ AI Enthusiastic and Programmer

- implementing neural networks to do classification on SVHN dataset
  - prepare data for feeding into neural network
  - built convolutional neural network as model
  - feed data to network and train with softball policy
- create a word cloud based of specific text data based on Natural Language Processing Methods
  - extract and prepare text data from a message platform like telegram
  - create model to output word cloud and feed text to that to get word cloud
- using {monte carlo or td error and tabular q-values} RL method to solve multi armed bandit problem
  - specify e-greedy policy and initialize tabular q-value based states and train {state or state-action} values based on {monte carlo or td error} method
- implementing uninformed and informed search algorithm (BFS, DFS, UCS, A*) for finding shortest path between origin node to target node in graphs and trees datastructures
- implementing couple of sorting and searching algorithms in python
- visualization of crossover and mutation operators proposed in NEAT algorithm (NeuroEvolution of Augmenting Topologies), a genetic algorithm for the generation of evolving artificial neural networks (a website application with frontend web developer tools)
- designing an Academic CV Website
- designing a CV with Latex
- create a telegram bot for anonymous messaging

Honors and Awards

- Being in top 5% in universities entrance exam for Bachelor of Science in the Country
- Rank 2 in the Class with more than 20 student in Term of GPA in Master of Science
- Getting Government Scholarship to Enter University in Master Of Science
- Study B.Sc and M.sc at Tabriz University (top 10 university in iran)

Skills

Programming

- Python
  - tensorflow
  - pytorch
  - numpy
  - gym

Web Development

- HTML5
- CSS3
- JavaScript
- Bootstrap
- Jquery

Database

- SQL
- MySQL

Others

- Office
- Arduino
- SolidWork
- MATLAB
- Docker
- Linux
- git
- latex

Certificates & Courses

- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Fundamentals of Reinforcement Learning
- Prediction and Control with Function Approximation
- The Complete Web Development Bootcamp
- Complete Python Course: Beginner to Advance by Ali Hejazi
- getting started with TensorFlow 2
- The Complete SQL Bootcamp
- Git Complete: The Definitive step-by-step guide to git
- Linux Administration Bootamp: Go from Begineer to Advanced
- Neural Networks and Deep Learning
- Programming Foundations: Fundamentals
- Sample-based Learning Methods
- A Complete Reinforcement Learning System
- Deep Neural Networks with PyTorch
- Docker Mastery: with Kubernetes + Swarm from Docker Capitan
- Customising your models with TensorFlow 2
- The Ultimate MySQL Bootecamp: Go from SQL Beginner to Advanced
- GIT: Advance Commands

References

1. **Behruz Koohestani,** Associate Professor of Computer Science, Tabriz University,
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2. **Behnam Dadashzadeh,** Associate Professor of Computer Science, Tabriz University,
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